SaiFE Findings Summary

Safe AI Framework for Speed-Limit Evaluation

14,711 usable segments scored	**6,729** priority segments scoring 50+	**131** critical-review segments	**250** time-based pilot candidates

What Problem SaiFE Solves

SaiFE helps transport agencies identify road segments where posted speed limits, actual operating speeds, road context, and geometry may create elevated safety concerns, especially for pedestrians, cyclists, and powered two-wheeler users. It is a transparent screening tool for deciding where governments should investigate first.

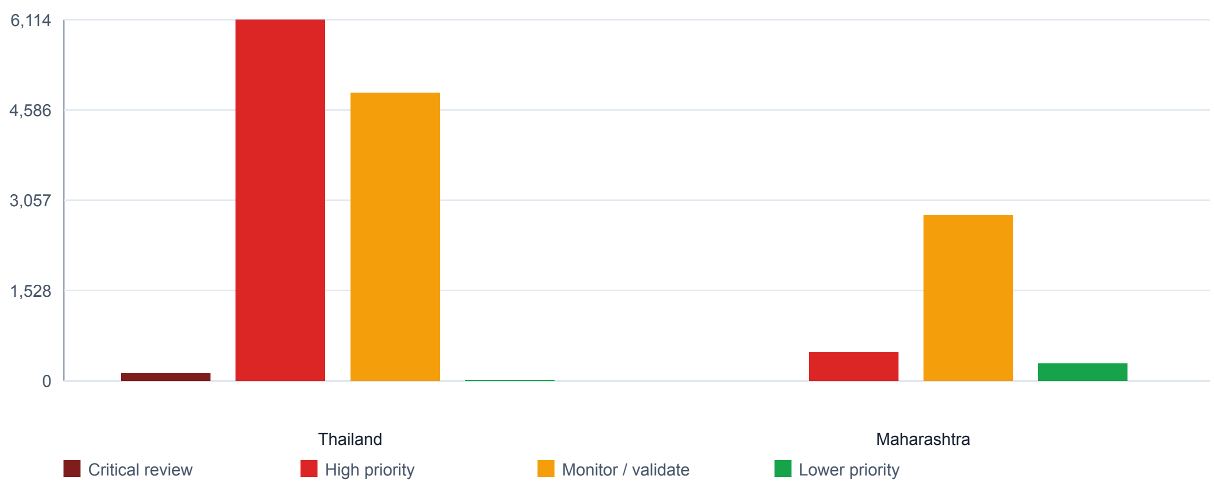
Method in Brief

The model uses the supplied Thailand and Maharashtra GeoJSON datasets. Each valid segment receives a 0-100 Speed Safety Score from six components: posted limit alignment, operating speed behavior, exposure proxy, vulnerable-user context, geometry, and data-quality review. Segments scoring 50 or above are treated as priority speed-unsafe segments for review.

Core Findings

Thailand produced 131 Critical-review segments and 6,114 High-priority segments. Maharashtra produced 484 High-priority segments. The strongest signals occur where high observed speeds combine with urban or major-road context, geometry concerns, and exposure proxies.

Priority classes by country

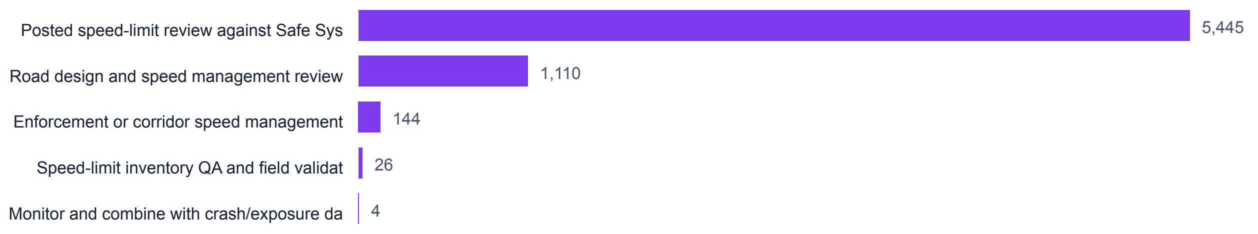


Segments by Speed Safety Score band

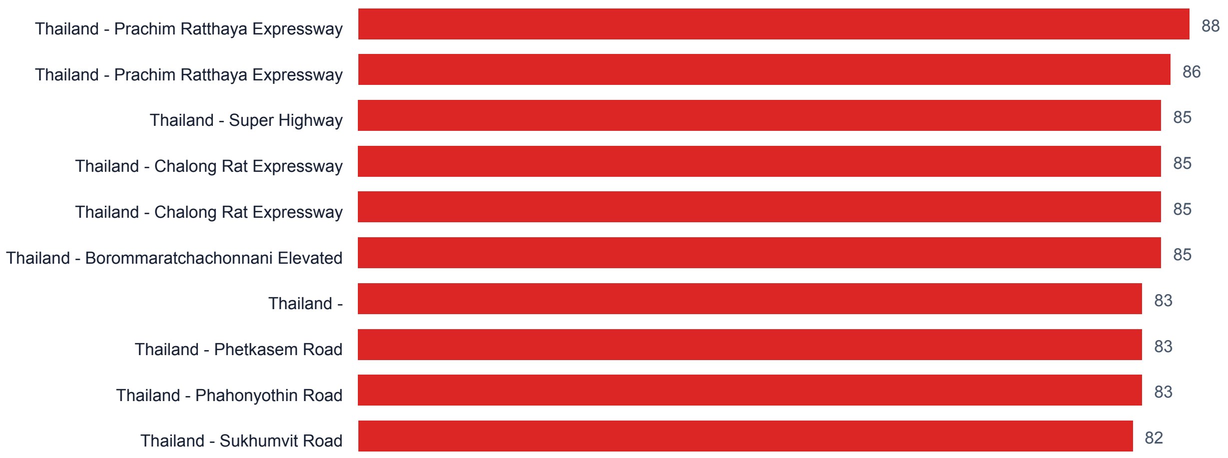
Policy-Relevant Outputs

SaiFE does not only rank locations. It assigns action categories that a ministry can use for speed-limit review, road-design review, corridor speed management, enforcement planning, QA, or monitoring.

Recommended action categories for priority segments



Top scored segments



Top Scored Segments

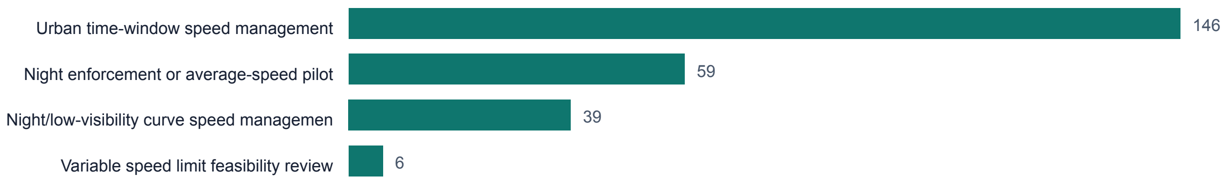
Country	Road	Class	Land use	Limit	F85	Score	Recommended action
Thailand	Prachim Ratthaya Expressway	motorway	URBAN	80.0	114.5	88	Road design and speed management review
Thailand	Prachim Ratthaya Expressway	motorway	URBAN	80.0	114.83333333	86	Road design and speed management review
Thailand	Super Highway	trunk	URBAN	80.0	95.5	85	Road design and speed management review
Thailand	Chalong Rat Expressway	motorway	URBAN	80.0	108.66666666	85	Road design and speed management review
Thailand	Chalong Rat Expressway	motorway	URBAN	80.0	107.11111111	85	Road design and speed management review
Thailand	Borommaratchachonnani Elevated Highway	primary	URBAN	80.0	99.85714285	85	Road design and speed management review
Thailand		trunk	URBAN	90.0	113.33333333	83	Road design and speed management review
Thailand	Phetkasem Road	trunk	URBAN	80.0	97.2	83	Road design and speed management review

Country	Road	Class	Land use	Limit	F85	Score	Recommended action
Thailand	Phahonyothin Road	trunk	URBAN	80.0	108.0	83	Road design and speed management review
Thailand	Sukhumvit Road	trunk	URBAN	80.0	95.0	82	Road design and speed management review
Thailand	Yantarakit Koson Road	trunk	URBAN	80.0	103.0	82	Road design and speed management review
Thailand	Phet Kasem Road	trunk	URBAN	80.0	91.5	82	Road design and speed management review

Time-Based Speed Management

The provided GeoJSON files do not contain timestamp, hour, or day/night fields, so SaiFE does not claim to prove day/night speeding patterns. Instead, it identifies where segment-hour data should be collected first for variable speed limits, night enforcement, school-zone windows, or dynamic speed-management pilots.

Time-based pilot candidate types



First-Wave Time-Based Pilot Candidates

Country	Road	Class	Score	Pilot type	Pilot score
Thailand	Prachim Ratthaya Expressway	motorway	88	Night/low-visibility curve speed management	116.59
Thailand	Prachim Ratthaya Expressway	motorway	86	Night/low-visibility curve speed management	114.55
Thailand	Chalong Rat Expressway	motorway	85	Night/low-visibility curve speed management	111.48
Thailand	Chalong Rat Expressway	motorway	85	Night/low-visibility curve speed management	110.41
Thailand	Udon Ratthaya Expressway	motorway	79	Night enforcement or average-speed pilot	107.54
Thailand	Borommaratchachonnani Elevated Highway	primary	85	Urban time-window speed management	106.43
Thailand	Buraphawithi Expressway	motorway	79	Night enforcement or average-speed pilot	106.25
Thailand	Phahonyothin Road	trunk	83	Urban time-window speed management	104.85

How a Ministry Could Use This

- Prioritize corridors for speed-limit review and Safe System alignment checks.
- Select locations for average-speed enforcement, fixed cameras, or speed feedback signs.
- Identify curved or high-speed roads where engineering treatment may be needed.
- Target field validation and speed-limit inventory QA.
- Choose first-wave corridors for segment-hour day/night speed analysis.

Validation and Limitations

> SaiFE is a screening and prioritization model, not a final crash-prediction model. The supplied data does not include crash outcomes, traffic volumes, vulnerable-user exposure, or time-of-day speed records. Final investment decisions should join the score with crash data, traffic counts, field audits, and local engineering review.

Replication

The method can be replicated in other countries using common road data: segment geometry, posted speed limit, road class, land use, observed speed statistics, and sample count. Countries with less data can start with posted limits, geometry, road class, and urban/rural context, then add speed observations and crash data as they become available.

Repository: https://github.com/tan-amos/adb-ai-for-safer-roads
Interactive map: https://tan-amos.github.io/adb-ai-for-safer-roads/